Lab Assignment 5: Overfitting and Regularization

Siddharth Narayan Mishra (123CS0189)

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Q1: Overfitting in Non-Linear Regression

Part A: Effect of Model Complexity

Polynomial regression models of degree $d \in \{1, 2, 3, 4, 5, 6\}$ were trained using gradient descent for 100 epochs. For low-degree models ($d = 1, 2$), both training and testing R^2 values were low, indicating underfitting due to insufficient model capacity.

As the polynomial degree increased, training R^2 consistently improved. However, after a certain degree (typically $d \geq 4$), the testing R^2 stopped improving and began to degrade while training R^2 remained high. This divergence between training and testing performance indicates the onset of overfitting. High-degree models fit noise in the training data, resulting in poor generalization.

Part B: Effect of L2 Regularization

For a fixed polynomial degree of $d = 6$, Ridge Regression models were trained with regularization strengths $\lambda \in [0, 1]$. Without regularization ($\lambda = 0$), the model exhibited overfitting, with high training R^2 but lower testing R^2.

As λ increased, testing R^2 improved initially, demonstrating better generalization due to penalization of large weights. For very large λ, both training and testing R^2 decreased, indicating underfitting. Thus, L2 regularization controls overfitting by constraining model complexity.

Q2: Overfitting in Non-Linear Binary Classification

Part A: Demonstration of Overfitting

Logistic Regression was trained on a non-linearly separable dataset for 100 epochs. Training accuracy increased steadily and reached high values, while testing accuracy saturated earlier and remained lower. The growing gap between training and testing accuracy across epochs indicates overfitting, as the model adapts closely to training data patterns.

Part B: Effect of L2 Regularization

L2-regularized Logistic Regression models were trained with $\lambda \in [0, 1]$. Moderate regularization improved testing accuracy by reducing variance and preventing large parameter values. Excessive regularization led to reduced accuracy due to underfitting. Hence, L2 regularization improves generalization by balancing bias and variance.

Conclusion

Overfitting occurs when model complexity exceeds the information content of the data. Increasing model capacity improves training performance but may harm generalization. Regularization techniques such as L2 penalty effectively mitigate overfitting by constraining model parameters and improving robustness on unseen data.